

Benzothiazoles (Mercapto Mix)



What is it?

Benzothiazoles are rubber accelerators. A rubber accelerator is a chemical used when making rubber. Rubber accelerators speed up the liquid to solid rubber transformation. Benzothiazoles can become a part of the final product in this process. Benzothiazoles may also be found in non-rubber items.

This is different from a latex allergy.

Mercapto mix may contain a combination 3 or 4 chemicals:

- N-cyclohexyl-2-benzothiazyl-sulfenamide
- Dibenzothiazyl disulfide (MBTS)
- Morpholinyl mercaptobenzothiazole (MOR)
- 2-mercaptobenzothiazole (MBT)

2-mercaptobenzothiazole (MBT) is also sometimes tested alone.

BENZOTHIAZOLES can also be known as:

- 2-Mercaptobenzothiazole
- 2-Benzothiazolethiol
- Mercaptobenzothiazole
- 2(3H)-Benzothiazolethione
- Benzothiazolethiol
- MBT
- Dermacid
- Benzothiazole-2-thiol
- Sulfadene
- N-cyclohexyl-2-benzothiazole-sulfenamide
- Dibenzothiazyl disulfide (2,2-Benzothiazyl disulfide, MBTS)
- Morpholinyl mercaptobenzothiazole (4-Morpholinyl-2-benzothiazyl disulfide, MOR)

Some Possible Uses of BENZOTHIAZOLES / Where it might be found:

Rubber items:

- Gloves (i.e. household, work, hospital: nitrile, latex, rubber)
- Rubber shoes, insoles, adhesives, linings, and canvas portions of different shoe types
- Clothing or swim wear with elastic
- Makeup applicator, rubber eyelash curler
- Medical devices (i.e. dialysis equipment, stethoscope, IV port, tourniquet, BP cuff, CPAP strapping, tubing, masks, etc.)
- Tires
- Toys, balls, sports equipment (i.e. handles, pads)
- Ear plugs, headphones
- Condom or diaphragm
- Rubber hip wader, gators
- Support stockings, non-slip waist bands
- Brassiere cups, underwear or panties
- Swimwear, swim cap, goggles, sauna suits
- Wetsuit, diving gear
- Rubber vibrator, rubber dropper
- Rubber pillow, mattress, or sheet
- Hairbrush handle, hair net
- Toothbrush handle, dentures, dental dams
- Baby bottle nipple, binky
- Balloons
- Tape, bunion pad
- Hot water bottle, urinal
- Braces and supports
- Rubber bands, iPod bands
- Pen handle, pencil grip, eraser
- Mouse pad, wrist support
- Glue, envelopes, stamps, rubber bank note counter
- Kitchen utensil rubber handle
- Rubber mat, carpet backing, underlay pad
- Upholstered furniture

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- Rubber cement
- Rubber hose, tubing, seals, cables
- Steering wheel, bike handle, fuel tank lining
- Auto hoses, gaskets, fan belts
- Shock absorber, springs, inner tubes
- Tool handles, duct tape, cords
- Weather stripping
- Safety apron
- Respiratory, gas mask
- Conveyor belt
- Synthetic resins in the textile/dye industry
- Eye pieces (i.e. on a microscope, binoculars)

Non rubber items:

- Anti-septics
- Repellant, anti-fungus/slime/pest/insect
- Glues, cements, cutting oil
- Greases
- Detergent
- Anti-freeze
- Soaps and shampoos
- Tick/flea powder and spray
- Photographic film emulsion

Other products that might react like BENZOTHIAZOLES:

- None

What should I do to avoid BENZOTHIAZOLES?

Avoid items that have benzothiazoles.

Use VINYL, non-latex gloves when you clean the house or wash dishes.

Wear heavy duty vinyl gloves to protect against workplace contact. Use fabric, leather, or plastic gloves if you need to touch dry rubber. If you think

that you have workplace exposure, consult the Material Safety Data Sheet (MSDS), or contact the manufacturer.

Elastic waistbands and underwear with elastic may contain rubber. Choose underwear that does not have an elastic waistband, or buy all-cotton clothing (cottonique.com, rawganique.com, decentexposures.com; make sure rubber-free).

Rubber may be in some forms of contraception. This includes condoms or diaphragms. There are condoms made of materials safe for those allergic to rubber or latex. The options for male condoms are: polyurethane (Trojan Supra) or lambskin (Trojan Naturalamb).

If you are benzothiazole sensitive and have foot dermatitis, it may be due to shoes containing rubber. Wear all leather shoes with no inner or outer sole, like moccasins. Molded plastic shoes or wooden clogs are an option. Contact a local shoe store and ask for a rubber-free shoe or contact an orthotic shop to have a custom made shoe. If this is not possible, remove the insoles. Insert insoles cut from piano felt, cork, or plastic. Limit sweating. Throw away socks that may have absorbed allergen.

Ask your provider for other options if your products or items may contain rubber.

What products may NOT have mercapto mix?

- Showa N-Dex Accelerator-Free nitrile gloves
- Vinyl, plastic, silicone, polyurethane, polyethylene, neoprene or acrylic gloves
- Honeywell Safety Silvershield/4H chemically resistant gloves

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- Examination gloves: Showa N-Dex
Accelerator-free nitrile gloves, Ansell
Micro-Touch NitraFree, SmartPractice
Reflection Sky Sensitive Nitrile
- Surgical gloves: Ansell GAMMEX Non-
Latex Neoprene Surgical glove, Medline
DermAssure Neoprene Surgical glove,
Mölnlycke Health Biogel NeoDerm Surgical
glove

In addition, the ACDS Contact Allergen Management Program® (CAMP®) can help manage contact dermatitis. It helps to find products that are safe to use by creating a list of products that do not contain one's allergens. For more information about CAMP®, please ask your doctor or visit www.contactderm.org.

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